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GenAI as an instant, readily available co-developer: redefining software design process for AI native hybrid work

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Abstract
The increasing adoption of hybrid and remote work has reshaped how software teams communicate, collaborate, and make design decisions. Our prior study showed that hybrid setups often result in missed discussions, fragmented understanding, and limited documentation of design rationale. At the same time, Generative AI (GenAI) tools–such as GitHub Copilot and ChatGPT–are becoming embedded in development workflows, providing support in code generation. While GenAI's role in software development is gaining attention, its impact on early-stage design ideation activities and collaboration in hybrid teams remains underexplored. This paper builds on earlier findings to propose a forward-looking vision of GenAI as an instant, readily available co-developer in hybrid software design. We revisit documented challenges from our earlier study on hybrid collaboration and outline how GenAI could address these issues by facilitating asynchronous participation, surfacing undocumented rationale, and preserving design continuity. We present a conceptual framing of GenAI-supported collaboration and propose a research agenda to guide future studies on integrating GenAI into hybrid settings.

Keywords Remote collaboration · Hybrid work · GenAI · Software design process

1 Introduction

The shift toward remote and hybrid work has fundamentally altered how software teams operate. While remote work has existed for decades (Olson 1983), it gained traction in software engineering through practices such as offshoring (Messenger and Ghosheh 2010), global software development (Herbsleb and Moitra 2001), and open

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source development (Crowston et al. 2008). The COVID-19 pandemic accelerated the normalization of remote and hybrid work models (Griffin 2021), although many organizations are now encouraging partial returns to the office (Telford 2023). Still, the notion of working from anywhere continues to influence workplace expectations (Sako 2021). These evolving work arrangements intersect with foundational socio-technical principles in software engineering. Conway's Law (Conti et al. 2009) and the theory of socio-technical congruence (Mauerer et al. 2021; Cataldo et al. 2008) underscore how communication structures shape the design of software systems. Agile frameworks such as Scrum (Schwaber 2004) emphasize autonomous, self-organizing teams, but in hybrid environments, collaboration practices are shaped by team-specific demands and practices rather than state-of-the-art methodologies (Santos et al. 2024). In this regard, development tools and communication technologies play a central role in shaping how collaboration unfolds across distributed environments (Kilamo et al. 2015).

Recently, the rise of generative AI (GenAI) tools – such as GitHub Copilot, ChatGPT, Amazon Q Developer, Google Gemini Code Assist, and CodeWhisperer – has introduced a new dimension to software development workflows (Matobobo et al. 2025; Mikkonen 2025; Mikkonen and Taivalsaari 2025). While primarily associated with code generation, these tools are increasingly recognized for their potential to support higher-level design activities, including architectural reasoning, ideation, and rationale documentation (Dhar et al. 2024; Díaz-Pace et al. 2024). In particular, distributed and hybrid settings–where synchronous communication is fragmented and design alignment is often difficult (Jackson et al. 2023)–GenAI can act as an instant collaborator, or, as put in (Hicks et al. 2024), the new developer. It can help bridge communication gaps, co-develop architectural decisions, and assist team members in maintaining shared understanding. Our prior work (Mikkonen et al. 2025) observed that hybrid setups often lead to fragmented participation, undocumented decisions, and missed opportunities for shared understanding. These limitations suggest an opportunity for rethinking how design collaboration unfolds – particularly with the emergence of GenAI collaborators that could help maintain continuity, surface rationale, and support alignment across time and space. Figure 1 presents a scenario of asynchronous collaboration across hybrid teams supported by GenAI. This example reflects a future collaboration model, where GenAI operates as a co-developer teammate–facilitating asynchronous ideation, preserving design rationale, and helping teams maintain shared understanding despite time and distance.

Building on our earlier study (Mikkonen et al. 2025), this paper outlines a design space for GenAI-supported collaboration in hybrid settings. We present challenges derived from previous empirical observations and conceptualize how GenAI could address them by acting as a co-developer that helps bridging gaps in communication, rationale, and alignment. While our earlier study empirically examined breakdowns in hybrid software design collaboration, this paper extends those findings by reconceptualizing the collaborative design process itself in the presence of GenAI by offering a rigorous, literature-grounded vision and structured research agenda, grounded in existing empirical findings and prior research, intended to guide future investigation and system design.

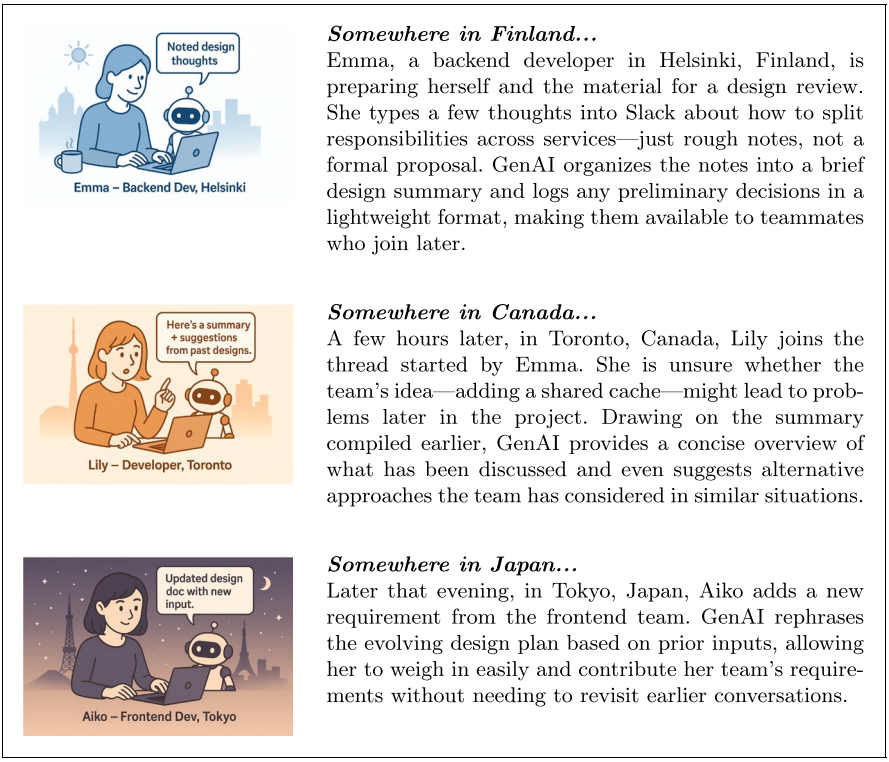


Fig. 1 GenAI as a co-developer across hybrid team: an illustrative scenario (AI-generated images)

The paper is structured as follows. Section 2 presents related literature and the motivation of the study. Section 4 details the proposed vision and Section 5 introduces research directions for understanding and shaping the role of GenAI in hybrid software design. Section 6 discusses the proposed vision and Section 7 concludes the paper.

2 Background and motivation

Software engineering is a well-established discipline that integrates socio-technical, cognitive, organizational, and collaborative processes to create meaningful software artifacts (Curtis et al. 1988). Different team roles, expertise, and group dynamics significantly shape project outcomes (Sonnentag 1998). Agile methodologies promote self-organizing teams (Moe et al. 2008; Hoda et al. 2012), which tend to thrive in co-located environments where informal interactions are easy to initiate and sustain (Strode et al. 2012). In practice, however, teams often adapt methodologies like Scrum to fit their specific needs (Hron and Obwegeser 2022), leading to deviations from existing processes and practices (Eloranta et al. 2016). Extending these insights to emerging software development practices, recent research on human–AI collaboration shows that collaboration outcomes are shaped by task structure, role clarity,

trust, and the allocation of responsibilities between human and non-human collaborators (Łabędzki et al. 2025; Siemon 2022). As software engineering practices continue to evolve, teams are also exploring new ways of collaborating in distributed and hybrid environments.

The increasing adoption of hybrid work, combining in-person and remote collaboration, has reshaped interaction patterns within software development teams. Many organizations use flexible models that integrate synchronous and asynchronous workflows (Santos et al. 2024; Kemell and Saarikallio 2023), influencing communication and collaboration practices during design and decision-making process (Jackson et al. 2023). Hybrid teams frequently use video conferencing, messaging platforms, and collaborative digital whiteboards to support coordination efforts (Yarmand et al. 2024; Leppänen et al. 2016). However, these tools often lack support for the spontaneous, co-creative exchanges typical in co-located settings (Kemell and Saarikallio 2023). As a result, the design process can become fragmented, with developers focusing on individual tasks instead of engaging in shared ideation and design planning (Yang et al. 2021). Existing research has explored various dimensions of hybrid software engineering–from communication and productivity (Espevik et al. 2011; Nguyen-Duc et al. 2024; Milfont et al. 2024) to team organization (Adzgauskaite et al. 2025). Existing tools for real-time interaction (e.g., collaborative code editors (Lautamäki et al. 2012)) focus on implementation instead of conceptual design. Similarly, practices such as pair programming promote collaboration (Nawrocki and Wojciechowski 2001); however, it often reinforces one-directional validation instead of equal participation in design. Related studies of hybrid and remote settings consistently report challenges in sustaining shared understanding, design rationale, and inclusive participation–particularly when collaboration unfolds asynchronously (Kemell and Saarikallio 2023; Jackson et al. 2023).

Generative AI (GenAI) has recently emerged as a promising practice in software engineering, supporting various stages of development (Zheng et al. 2024). Tools such as GitHub Copilot, ChatGPT, and other Large Language Models (LLM) assist not only with code generation but also with higher-order tasks such as explaining APIs, synthesizing documentation, and providing architectural suggestions (White et al. 2024). These capabilities create new opportunities for collaborative design activities, particularly in hybrid and distributed teams, where maintaining continuity, shared ideation, and rationale capture is challenging with traditional tools. A recent multivocal literature review by Esposito et al. (2025) identified gaps in the use of GenAI for collaborative design, especially during early-stage ideation in hybrid settings. A recent systematic review on human–AI collaboration highlighted recurring challenges such as over-reliance, trust calibration, and role ambiguity, indicating that AI collaborators can both support and undermine team outcomes depending on how they are integrated into work practices (Łabędzki et al. 2025). These findings underscore the need for targeted research on GenAI's role in shaping collaborative software design in hybrid environments. Understanding the impact of hybrid work on team coordination, communication, and collaborative decision-making provides a foundation for situating GenAI contributions in real-world settings. Existing studies have examined these dimensions in remote and hybrid environments (Espevik et al. 2011; Kemell and Saarikallio 2023; Nguyen-Duc et al. 2024), highlighting breakdowns and

adaptations that emerge in practice. Other studies explore how hybrid work affects collaboration, productivity, communication, and team structure. Espevik et al. (2011) examine how communication patterns evolve under hybrid constraints, Nguyen-Duc et al. (2024) and Milfont et al. (2024) investigate organizational practices and managerial approaches, and Kemell and Saarikallio (2023) and Adzgauskaite et al. (2025) study hybrid team organization and cross-role coordination. Collectively, these studies provide important insights into how software teams adjust to hybrid, remote, and in-office settings, highlighting the challenges and opportunities that GenAI could help address.

The emergence of GenAI tools introduces a significant shift in how developers engage with design-related tasks (Mikkonen and Taivalsaari 2025). Our previous studies have shown that developers increasingly use external tools initially Stack Overflow (Mäkitalo et al. 2020), and more recently GenAI systems to seek solutions or feedback, often bypassing traditional co-developer interactions (Burtch et al. 2024). This shift suggests that our earlier hypothesis–that developers' ability to interact with each other strongly shapes software design outcomes (Mikkonen et al. 2025)–must now consider new types of "teammates" in the form of GenAI collaborators. Recent empirical studies highlight that developers now use GenAI systems as collaborative partners, while developers themselves remain the final authority on complex reasoning, security, and architectural decisions (Hamza et al. 2024).

Designing and implementing software systems involves complex, iterative decision-making that is sensitive to team workflows, tool support, and the increasing integration of AI collaborators. In hybrid teams, physical and temporal distance adds challenges to synchronous collaboration and shared ideation. GenAI offers new possibilities while raising concerns about over-reliance and reduced human oversight. A recent study (Kosmyna et al. 2025) presented that delegating cognitive tasks to GenAI can reduce cognitive engagement, highlighting the importance of understanding the trade-offs involved. Recent research on AI-based teammates further shows that perceptions of AI as team members can influence trust, satisfaction, and conflict, reinforcing the need for deliberate role assignment and governance (Dennis et al. 2023; Siemon 2022).

As GenAI tools become integrated into collaborative design practices, it is essential to examine how developers interact with them and how these interactions differ from traditional human-to-human collaboration. Addressing these challenges requires both technical innovation and deliberate process design to assign clear roles and responsibilities to human and AI collaborators.

3 Research approach

This paper adopts a literature-grounded, empirically informed research approach to articulate a vision of GenAI as a co-developer in hybrid software design. To improve transparency and traceability, we structure this approach into a sequence of empirically grounded and literature-informed steps, as shown in Fig. 2. First, we revisited the empirical challenges identified in our prior study of software design processes in co-located, remote, and hybrid teams (Mikkonen et al. 2025). These challenges

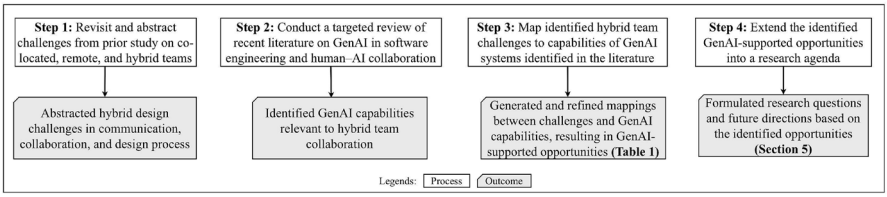


Fig. 2 Overview of the structured process used to identify GenAI-supported opportunities in hybrid software design

included fragmented collaboration, undocumented design rationale, and misalignment, as remote participants often struggled to follow in-room discussions. From this analysis, we abstracted three recurring categories of challenges: communication and documentation, collaboration and alignment, and continuity of the design process. To address these challenges, we conducted a targeted review of recent research on GenAI in software engineering and human–AI collaboration. The review focused on studies published between 2020 and 2025 in major international software engineering and human–computer interaction conferences proceedings and journals. We selected empirical studies, systematic reviews, and industrial reports relevant to collaborative design, hybrid work practices, and AI-supported coordination, with the aim of understanding how GenAI can be integrated into collaborative workflows. Recent research shows that GenAI systems are increasingly integrated as meeting assistants, conversational partners, and workflow agents to support collaborative work (Łabędzki et al. 2025; Boyd 2025). Based on the challenge categories derived from our prior study and the capabilities of GenAI systems discussed in the literature, we constructed a mapping between hybrid design challenges and potential GenAI-supported solutions. To construct this synthesis, we followed a three-step mapping procedure: *(1)* aligning challenge categories from our prior study with capabilities of GenAI systems identified in the literature; *(2)* generating mappings between identified challenges and the capabilities of GenAI systems; and *(3)* iteratively refining these mappings based on insights from the literature and discussion among the authors. The resulting mappings are summarized in Table 1, which presents GenAI-supported opportunities for addressing challenges in hybrid software design across the identified categories.

4 GenAI as co-developer in hybrid work: a vision

In our prior study (Mikkonen et al. 2025), we analyzed how the software design process unfolds in co-located, remote, and hybrid teams. The findings revealed that different work setups influence the form and frequency of design interactions, as well as the clarity, documentation, and ownership of architectural decisions. In hybrid teams in particular, remote participants often missed in-room discussions, leading to fragmented understanding of the design context and reduced participation in architectural decision-making. Across all setups, we observed that teams tended to split work early and shift quickly toward implementation, limiting opportunities for collective design exploration and rationale building. While challenges in hybrid collaboration

Table 1 Challenges in hybrid software design (based on Mikkonen et al. (2025)) and illustrative GenAI-supported approaches

Challenge	How GenAI may address the challenge	Example
Communication and documentation		
Remote participants miss in-room discussions	Provides summaries of design discussions and bridges gaps for asynchronous participants	GenAI summarizes key decisions and unresolved points, making them accessible to remote team members
Design rationale remains undocumented	Captures rationale from informal discussions or code comments and stores them alongside design artifacts	A team member writes a reasoning comment during a Slack exchange, which GenAI logs as part of the evolving design history
Collaboration and alignment		
Team members split work early without shared design	Encourages idea convergence by proposing alternative solutions and prompting team-level discussion	GenAI notices parallel implementation tasks and prompts a quick alignment check on design assumptions
Remote member(s) hesitate to interrupt synchronous sessions	Serves as a passive contributor that enables quiet brainstorming and asynchronous questioning	A remote developer types a question into the design chat; GenAI provides clarifications and surfaces related prior discussions
Design process		
Lack of clarity around architectural decisions	Maintains linked summaries of architectural choices, rationale, and alternatives	GenAI retrieves a previous architecture discussion and explains why a particular architecture was selected
Difficulty revisiting or continuing partial designs	Stores intermediate design artifacts and prompts teams to resume unfinished threads	After a break, GenAI brings forward prior diagrams, suggestions, and rationale to help continue the discussion

have been observed in human-to-human teams (Mikkonen et al. 2025), the recent rise of GenAI introduces new opportunities to rethink participation in the software design process. In this paper, we envision GenAI as a co-developer whose primary contribution lies in mediating design interactions, preserving design knowledge, and supporting alignment across time and space. This includes making design discussions, decisions, and rationale accessible to distributed team members. While prior work has explored AI-supported collaboration through concepts such as AI assistants, copilots, and AI teammates, our vision of GenAI as a co-developer emphasizes a distinct role in the software design process. Instead of focusing on task-level assistance (e.g., code generation or completion) or fully autonomous team participation, we conceptualize GenAI as a participant that mediates design interactions, supports the articulation and persistence of design rationale, and facilitates alignment across

distributed team members. This perspective shifts the focus from productivity augmentation or autonomy toward supporting collective design reasoning and shared understanding in hybrid settings, which remains underexplored in existing literature.

Recent literature shows that prompt-based interactions with large language models can support developers in understanding and maintaining software written by others, suggesting a concrete role for GenAI in onboarding and knowledge transfer tasks (Uusnäkki et al. 2025). It further highlights the increasing use of GenAI in software development activities (Ahmad et al. 2023; Dhar et al. 2024; Díaz-Pace et al. 2024), yet its role in supporting collaborative design, particularly in hybrid settings, remains underexplored (Łabędzki et al. 2025). At the same time, recent research on AI-based teammates indicates that unclear or overly agentic AI roles can undermine trust, satisfaction, and coordination within teams (Dennis et al. 2023). To address these challenges, our vision deliberately positions GenAI as a collaborator that augments human participation rather than replacing it, emphasizing clear role definition, appropriate task boundaries, and deliberate integration into team workflows to ensure that AI enhances collaboration while maintaining human oversight. Figure 3 presents a conceptual interaction model illustrating how GenAI may participate as a co-developer in hybrid software design settings. The model highlights key interaction patterns, boundaries, and control points between developers, GenAI assistants, and a shared collaboration space. It serves as a reference for understanding how GenAI can support design exploration, documentation, and alignment by summarizing discussions, capturing design rationale, and retrieving prior design context.

Building on the interaction model, Table 1 reframes challenges observed in hybrid software design (Mikkonen et al. 2025) by illustrating how GenAI could participate in the collaborative software design process. The mappings presented in the Table 1 are informed by capabilities of GenAI systems discussed in the literature and reflect

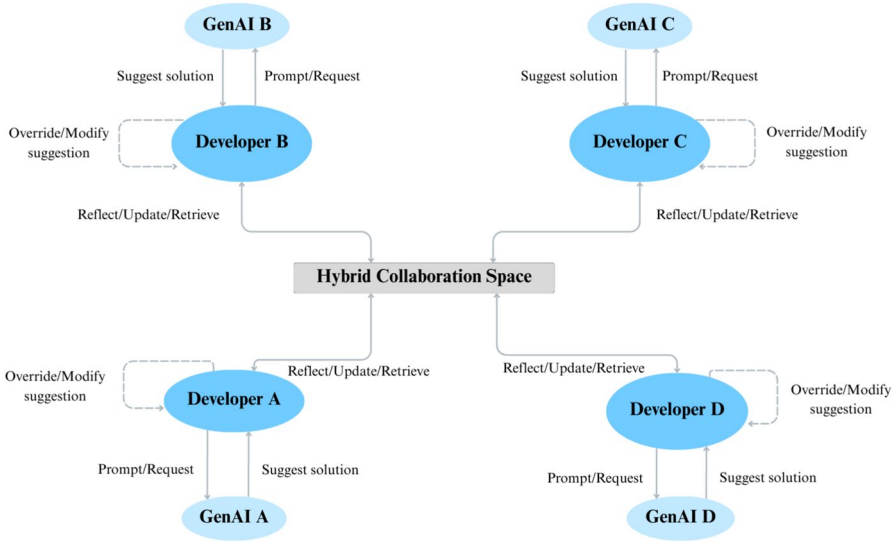


Fig. 3 Conceptual interaction model of GenAI as a co-developer in hybrid software teams, showing developer–GenAI interactions and reflection into a shared collaboration space

how such capabilities may address specific breakdowns observed in hybrid design settings. These scenarios serve as conceptual examples of how GenAI might mediate collaborative design work while keeping design knowledge visible and open to discussion. The following section outlines a research agenda to guide future work on the role of GenAI in hybrid software design.

5 Research agenda for GenAI-supported design

The vision outlined in this paper raises several research directions for understanding and shaping the role of GenAI in hybrid software design. These directions build on our prior empirical findings on collaboration breakdowns in co-located, remote, and hybrid teams (Mikkonen et al. 2025), and respond to growing interest in structured and theory-informed research on human–AI collaboration (Łabędzki et al. 2025). We outline a set of research questions below to guide future work. These research questions are intended to support future empirical studies of how GenAI-supported mechanisms influence communication, alignment, and decision-making in hybrid teams.

RQ1: How does the use of GenAI as a co-developer reshape collaborative design practices across co-located, remote, and hybrid teams?

While GenAI tools are increasingly used across software development activities (Ahmad et al. 2023; Dhar et al. 2024; Díaz-Pace et al. 2024), their impact on collaborative design practices likely varies depending on team configuration and work arrangement. Research on LLM-based meeting assistants and collaborative AI systems suggest that GenAI can support shared understanding and asynchronous participation, but its effects largely depend on how it is embedded in team workflows (Cabrero-Daniel et al. 2024). Further research is needed to explore how GenAI as co-developer supports summarization, ideation, and rationale capture, and how these capabilities influence participation, alignment, and decision-making in co-located, remote, and hybrid teams.

RQ2: How does GenAI as a co-developer reshape the use of software tools and workflow support in hybrid settings?

GenAI tools are increasingly integrated into communication platforms, IDEs, and workflow systems, potentially reconfiguring how developers document decisions, coordinate work, and engage in team discussions. Research on human–AI collaboration in software engineering highlights that introducing GenAI into existing tools can alter coordination practices and shift attention away from traditional documentation and communication mechanisms (Hamza et al. 2024; Mohamed et al. 2024). Industrial visions of agentic workflows further position GenAI as an active participant within collaborative work environments, accelerating these changes (Boyd 2025). Further research is needed to explore how GenAI as a co-developer complements, replaces, or reshapes existing tools and practices, and how hybrid teams adapt their workflows in response.

RQ3: How should design responsibilities be allocated between human developers and GenAI as a collaborator in hybrid teams?

As GenAI tools become increasingly integrated into software development workflows, particularly through platforms such as ChatGPT and GitHub Copilot, organi-

zations are exploring how these systems can support collaborative design and coding activities (Boyd 2025). This raises important questions about how design responsibilities should be allocated between human developers and GenAI collaborators. Prior research on AI as team members suggests that unclear role definitions or overly agentic AI behavior can affect trust, satisfaction, and team coordination (Dennis et al. 2023; Siemon 2022). Future research is needed to examine when and how developers accept, adapt, or override AI-generated suggestions, and how responsibility allocation influences accountability, trust, and collaboration in hybrid settings.

While these research questions focus on the potential of GenAI as a co-developer, they also explicitly acknowledge the risks and trade-offs associated with introducing AI collaborators into design work. Prior research on human–AI collaboration highlights concerns around over-reliance, reduced critical engagement, and role ambiguity, particularly when AI systems are perceived as authoritative or autonomous contributors (Cabrero-Daniel et al. 2024; Dennis et al. 2023; Siemon 2022). In hybrid teams, where shared understanding and accountability are already fragile, such risks may be amplified. Hence, addressing these questions requires examining when GenAI supports collaboration, and also when it may hinder alignment, suppress dissent, or obscure responsibility.

Together, these research questions frame a research agenda that treats GenAI-supported design as a socio-technical phenomenon. The agenda emphasizes understanding how, when, and under what conditions GenAI reshapes collaborative design practices in hybrid teams. Addressing these questions will require empirical engagement with software practitioners working across different development settings. Mixed-method approaches–such as surveys, interviews, focus groups, and team observations–can be used to study design interactions, decision-making processes, and evolving team dynamics. The following section discusses the implications of this agenda, outlining key challenges, limitations, and considerations for responsibly integrating GenAI as a co-developer in hybrid software design.

6 Discussion

In this paper, we extend our earlier analysis of human-to-human collaboration challenges across co-located, remote, and hybrid teams (Mikkonen et al. 2025) by focusing specifically on how these challenges may evolve in the presence of GenAI. While the proposed vision and research agenda highlight promising directions for GenAI-supported design, it is also important to reflect on the limitations and risks that may constrain responsible adoption in hybrid software design settings (Esposito et al. 2025). Table 2 summarizes potential limitations of GenAI-supported design, mapping them to recurring hybrid design challenges and the relevant research questions needed to address them.

Limited awareness of architectural and team context. Current GenAI systems often operate with limited awareness of evolving system architectures, prior design decisions, and team-specific constraints. As a result, AI-generated summaries may lack alignment with project history or organizational context (White et al. 2024). In hybrid teams, where access to design discussions and their underlying rationale is

Table 2 Mapping of limitations in GenAI-supported design to hybrid design challenges and relevant research questions to address them

GenAI Limitation	Related Challenge(s) from Table 1	Relevant Research Question(s)
Limited awareness of architectural and team context	Lack of clarity in architectural decisions; design rationale remains undocumented	RQ1, RQ3
Fragmented workflow integration	Remote participants miss in-room discussions; remote member(s) hesitate to interrupt synchronous sessions; difficulty revisiting or continuing partial designs	RQ2
Excessive reliance on GenAI and diminished human engagement	Team members split work early without shared design; weakened shared understanding	RQ1, RQ3
Privacy, accountability, and organizational risks	-	RQ3
Developer preparedness and GenAI literacy	-	RQ2, RQ3

uneven, such limitations can undermine shared understanding when later participants rely on incomplete or outdated summaries. This highlights a fundamental tension in positioning GenAI as a co-developer (RQ1): while it can help surface and preserve design knowledge, it may also reinforce partial or incomplete interpretations unless mechanisms for validation and contextual grounding, and responsibility allocation (RQ3) are in place.

Fragmented workflow integration. GenAI tools are increasingly embedded across development tools, communication platforms, and collaboration environments; however, these integrations often remain fragmented at the team level. As a result, GenAI support is unevenly distributed across meetings, messaging tools, design artifacts, and code repositories, limiting its ability to support collective design work, particularly in hybrid settings where collaboration spans multiple tools, time zones, and participation modes (Mohamed et al. 2024). Without tighter workflow integration, GenAI risks supporting individual productivity more than team-level alignment, continuity, or participation (RQ2).

Excessive reliance on GenAI and diminished human engagement. A central concern in human–AI collaboration research is the risk of over-reliance on AI-generated outputs. In hybrid settings, excessive trust in GenAI summaries, suggestions, or alternatives may limit critical discussion and reduce team collaboration. Recent studies show that delegating cognitive work to GenAI can lower human engagement and diminish reflective reasoning (Kosmyna et al. 2025). In hybrid teams, this dynamic may further weaken shared ownership or reinforce early design decisions without sufficient team discussion. Research on AI teammates also indicates that such effects can influence trust, satisfaction, and overall collaboration quality (Dennis et al. 2023). These concerns underscore the need to examine how GenAI participation reshapes collaboration norms and responsibility allocation (RQ1, RQ3). In such cases, GenAI

may unintentionally narrow the design space by stabilizing early interpretations instead of supporting continued ideation.

Privacy, accountability, and organizational risks. Integrating GenAI into collaborative workflows raises concerns around data privacy, intellectual property, and accountability, particularly in safety-critical domains. In hybrid environments, where design decisions are distributed across tools and interactions, introducing GenAI as a co-developer may further complicate traceability and obscure responsibility. This highlights the need for transparent governance structures that clearly define human and AI responsibilities and ensure adequate human oversight to mitigate legal, ethical, and organizational risks (Kuhail et al. 2024). These concerns are critical when GenAI contributions influence architectural decisions, for which accountability remains firmly human but may become blurred in AI-mediated workflows. This raises questions about how responsibility for design decisions is negotiated between human developers and GenAI collaborators (RQ3).

Developer preparedness and GenAI literacy. Effective collaboration with GenAI requires a level of literacy that many developers may not yet possess. This includes understanding when to trust or challenge AI-generated suggestions, interpreting outputs critically, and integrating AI contributions into team discussions. Recent studies indicate that many developers lack training in how to collaborate responsibly with AI systems (Kuhail et al. 2024). As a result, GenAI risks amplifying existing inequalities in participation and influence within hybrid teams, particularly when AI-mediated contributions are perceived as more authoritative than human input (RQ2, RQ3).

These limitations underscore the need to approach GenAI-supported design as a socio-technical challenge, not only a technical one. While GenAI provides new opportunities to support collaboration in hybrid settings, its effectiveness depends on how it is embedded into team workflows, how responsibilities are allocated between human developers and AI collaborators, and how design knowledge is validated and governed. Instead of replacing human expertise, GenAI should be used as a co-developer to support reflective, traceable, and team-oriented design practices that preserve developer agency and shared accountability. Addressing these challenges requires empirical research to examine when GenAI supports collaboration and when it may hinder alignment, suppress dissent, or obscure responsibility in hybrid teams.

7 Conclusions

Hybrid work has fundamentally reshaped how software design unfolds, often introducing persistent challenges around shared understanding, participation, and the continuity of design rationale. Building on our earlier empirical analysis of collaboration across co-located, remote, and hybrid teams (Mikkonen et al. 2025), this paper examined how these challenges may evolve in the presence of GenAI. Rather than focusing on GenAI solely as a productivity tool, we conceptualized it as a co-developer that participates in collaborative design work by mediating interactions, preserving design knowledge, and supporting alignment across time and space.

This paper makes three main contributions. First, we articulate a vision of GenAI as a co-developer in hybrid software design, grounded in prior empirical findings and

recent research on human–AI collaboration. Second, we propose a structured research agenda that outlines key research questions for understanding how GenAI reshapes collaborative design practices, workflow integration, and responsibility allocation in hybrid teams. Third, we analyze potential limitations of using GenAI as a co-developer in hybrid settings and discuss how these limitations can be addressed through the research agenda proposed in this paper. While this study does not present an empirical evaluation or technical artifact, it provides a rigorous, literature-grounded foundation for future empirical and design-oriented research on GenAI-supported collaborative software design. As GenAI becomes increasingly embedded in development environments, understanding its role as a collaborator is essential for shaping future software engineering practices.

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Author Contributions Adil, Fronza, and Mikkonen wrote the main manuscript text Ihantola, and Iaccarino wrote small portion of text in Section 1 and 2 All authors reviewed the manuscript

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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